

# Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 23 papers, 2069–2082 BS

## 4. Association Analysis

10 hrs · in 23/23 papers · 5 topics, 3 TOP tier · **397 marks, 20.5% of the archive** · **157 numerical (40%) + 240 theory** · **heaviest numericals in the syllabus** → Ch 4 companion.

### Weight of the chapter:

- FP-Growth / FP-Tree **TOP 17/23** → **2nd most-asked topic in the whole subject.**
- Frequent Itemset & Apriori **TOP 14/23** → almost always a full numerical.
- Basics (support, confidence, market basket) **TOP 10/23**
- Sequential / Subgraph / Infrequent patterns **HOT 5/23** · Categorical attributes **PIN 3/23**

### 4.1 Basics and Algorithms

**What is association analysis? Explain its importance in market-basket analysis.** **2+5 10 2+6**

**TOP 10/23** 82 Bh, 80 Ba, 75 Ch, 75 Ash, 74 Ch, 73 Ch, 72 Ch, 72 Ka, 70 Ch, 70 Asa

#### Definitions

- **Itemset**: a set of 1 or more items.  $k$ -itemset = contains  $k$  items.
- **Transaction  $T$** : a set of items bought together.  $D$  = the whole database.
- **Association rule**: an implication  $A \Rightarrow B$  where  $A \subset I$ ,  $B \subset I$ , and  $A \cap B = \emptyset$ .
- **Association analysis**: finding all rules that satisfy a minimum support and a minimum confidence.

#### The 2 measures (learn both definitions exactly)

$\sigma(X)$  = support count of  $X$  (how many transactions contain it),  $|D|$  = total transactions.

$$\text{sup}(A \Rightarrow B) = P(A \cup B) = \frac{\sigma(A \cup B)}{|D|}$$

$$\text{conf}(A \Rightarrow B) = P(B|A) = \frac{\sigma(A \cup B)}{\sigma(A)}$$

- **Support** → how *frequent*. Filters out rules with too little evidence.
- **Confidence** → how *reliable*. Strength of the implication.
- A rule is **strong** if it meets *both* thresholds.
- **Support is symmetric** ( $A \cup B$ ), **confidence is not** ( $A \Rightarrow B \neq B \Rightarrow A$ ).

#### Market basket analysis 75 Ch short note

- Analyses which items customers put in the basket *together*.
- Classic result:  $\{\text{nappies}\} \Rightarrow \{\text{beer}\}$ .
- **Uses**: shelf layout, cross-selling, product bundling, catalogue design, promotion planning, recommendation (“customers who bought this also bought...”).
- **Actionable both ways**: put them together to increase convenience, or far apart to force the customer past other shelves.

#### 2-step process (every algorithm does this)

- 1. Find all frequent itemsets**: those with support  $\geq \text{min\_sup}$ . **This is the expensive step** and is what Apriori / FP-Growth optimise.
- 2. Generate strong rules** from them: for each frequent itemset  $l$ , take every non-empty proper subset  $s$  and output  $s \Rightarrow (l - s)$  if  $\frac{\text{sup\_count}(l)}{\text{sup\_count}(s)} \geq \text{min\_conf}$ .
- 3.** A  $k$ -itemset yields  $2^k - 2$  candidate rules.

#### Applications beyond retail 80 Ba asks where it is applicable

- Web usage / click-stream mining, medical symptom-disease co-occurrence, **gene co-occurrence** (Ch 1), fraud detection, intrusion detection, text mining.

## Pattern evaluation: why support-confidence is not enough 8

70 Ch “Why is pattern evaluation important? Explain the statistical based measures used for measuring interestingness.” **Full worked example in the companion.**

### The problem

- A rule can clear both thresholds and still be **completely misleading**.
- **Classic case:** 10,000 txns, 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.
  - $support = 4000/10000 = 40\%$  (passes)
  - $confidence(game \Rightarrow video) = 4000/6000 = 66\%$  (passes)
  - But  $P(video) = 75\%$  on its own, **higher than 66%**.
  - Buying a game makes you *less* likely to buy a video. The “strong” rule is **negatively correlated**.
- Cause: **confidence ignores the support of the consequent**.

### Correlation measures verified/added

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{conf(A \Rightarrow B)}{sup(B)}$$

- $> 1$  positively correlated ·  $< 1$  negative ·  $= 1$  independent.
- Above:  $lift = 0.40/(0.60 \times 0.75) = 0.89 < 1 \rightarrow$  caught.

$$\chi^2 = \sum \frac{(observed - expected)^2}{expected}$$

- Above:  $\chi^2 = 555.6$ , and observed  $<$  expected  $\rightarrow$  negative correlation.

### 4 null-invariant measures

- $all\_conf(A, B) = \frac{sup(A \cup B)}{\max\{sup(A), sup(B)\}} = \min\{P(A|B), P(B|A)\}$
- $max\_conf(A, B) = \max\{P(A|B), P(B|A)\}$
- $Kulc(A, B) = \frac{1}{2}(P(A|B) + P(B|A))$
- $cosine(A, B) = \frac{sup(A \cup B)}{\sqrt{sup(A) \times sup(B)}}$
- **Null-invariant** = unaffected by the number of transactions containing *neither* item. **lift** and  $\chi^2$  are **NOT null-invariant**, so they distort on sparse data.
- Use **Kulc** + **imbalance ratio** together for the clearest reading.

## Why a 100 % confidence rule is uninteresting, and why 99 % is 4

80 Ba “Generally we are more interested in rules with high confidence. However often we will not be interested in rules with a confidence of 100 %. Why? Then specifically explain why rules with 99 % confidence may be interesting.” verified/added

### 100 %: true by construction, so it teaches nothing

- $conf(A \Rightarrow B) = 100\%$  means **every** transaction holding *A* also holds *B*, no exceptions anywhere in *D*.
- Almost always that is a **property of the schema, not of the customers**  $\rightarrow$  already known before mining started:
  - **Definitional:**  $\{pregnant\} \Rightarrow \{female\}$ ,  $\{Corolla\} \Rightarrow \{car\}$ .
  - **Taxonomic:**  $\{Kathmandu\} \Rightarrow \{Nepal\}$ , item  $\rightarrow$  its own category.
  - **Subset artefact:**  $\{milk, bread\} \Rightarrow \{milk\}$ . Consequent already sits inside the antecedent’s itemset.
  - **Bundled by a business rule:** free gift always shipped with the product, so the pair can never come apart.
- **No headroom, so no action.** You cannot lift sales of *B* by promoting *A*: everyone who takes *A* takes *B* already. Cross-selling, shelf layout and recommendation all have nothing to work with.
- Such rules are also generated in bulk (every frequent itemset spawns its own subset rules) and **bury the genuine findings**. Fix: a maximum support threshold, or drop rules whose consequent is implied by the antecedent.

### 99 %: real regularity, and the 1 % is the payload

- A definitional rule cannot have exceptions. So **99 % proves the dependency is empirical**, discovered from behaviour rather than built into the data dictionary.
- It is still near-deterministic  $\rightarrow$  strong enough to predict on, unlike a 60 % rule.
- **The interest is in the 1 % that breaks it**, and it reads two ways. Give both:
  1. **The exceptions are wrong**  $\rightarrow$  candidate **data-quality defects**: mis-keyed records, typos, a faulty sensor, a duplicated ID. Near-100 % rules are used exactly this way to discover integrity constraints and clean a warehouse (Ch 2).
  2. **The exceptions are real**  $\rightarrow$  candidate **anomalies**: fraud, intrusion, a defective batch, a customer segment behaving unlike every other. This is the association-rule route into anomaly detection (Ch 6).
- Practical value too: 1 % is small but non-empty, so the customers who take *A* without *B* are a **precisely targetable list**.
- **One line worth memorising:** a 100 % rule tells you about the *schema*, a 99 % rule tells you about the *world*, and its exceptions are where the interesting cases live.

## 4.2 Frequent Itemset Pattern & Apriori Principle

Explain the Apriori algorithm / principle with example. 2+6 10 8

**TOP 14/23** 82 Bh, 81 Bh, 80 Bh, 80 Ba, 79 Bh, 79 Ba, 76 Ch, 76 Ash, 74 Ash, 73 Ch, 71 Ch, 71 Shr, 70 Ch, 69 Ch **Numerical in 7 of those** → companion.

The Apriori property (state this, it is the answer)

- All non-empty subsets of a frequent itemset must also be frequent.
- Contrapositive, and the one actually used: **if an itemset is infrequent, all of its supersets are infrequent** → prune them without counting.
- This is **anti-monotonicity**: support never increases as an itemset grows.
- **Effect**: turns a  $2^n$  brute-force search into something tractable. 71 Ch asks exactly this, 10.

Brute force, for contrast

- $n$  items →  $2^n - 1$  candidate itemsets, each needing a database scan.
- Apriori prunes the lattice level by level instead.

Algorithm

1. Scan  $D$ , count each item → keep those  $\geq \text{min\_sup}$  →  $L_1$ .
2. Repeat for  $k = 2, 3, \dots$  until  $L_{k-1}$  is empty:
  - (a) **Join**:  $C_k = L_{k-1} \bowtie L_{k-1}$ . Join 2 itemsets only if their **first  $k - 2$  items are identical** (keeps items sorted, avoids duplicates).
  - (b) **Prune**: drop any candidate having any  $(k - 1)$ -subset not in  $L_{k-1}$ .
  - (c) **Count**: scan  $D$  once, count the survivors.
  - (d)  $L_k =$  candidates meeting min\_sup.
3. Answer =  $\bigcup_k L_k$ .

Then generate rules

- For each frequent  $l$ , for each non-empty proper subset  $s$ : output  $s \Rightarrow (l - s)$  if  $\frac{\text{sup}(l)}{\text{sup}(s)} \geq \text{min\_conf}$ .
- **When no rule can exist at any confidence** 79 Ba 3 **verified/added**: a rule needs a *proper* non-empty subset, so it needs a frequent itemset of size  $\geq 2$ . If  $L_2 = \emptyset$  (every frequent itemset is a single item), then **no rule exists for any confidence threshold, including 0 %** → lowering min\_conf cannot help, only lowering min\_sup can. Check  $L_2$  before answering, never the confidence value.

Advantages

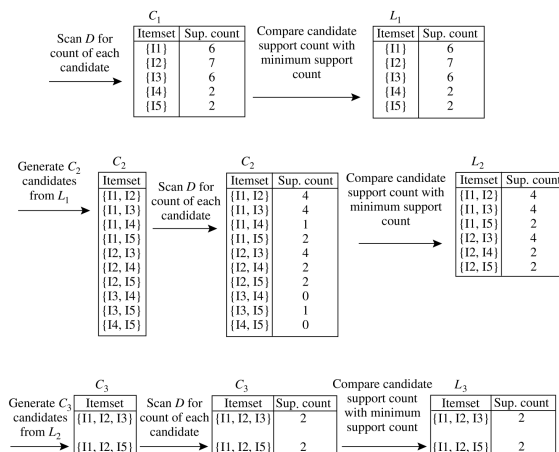
- Simple, easy to implement and to explain.
- Uses a level-wise search that is easy to parallelise.
- The pruning is **provably safe** (no frequent itemset is ever lost).

Disadvantages (the setup for FP-Growth)

- **Generates a huge number of candidates**.  $10^4$  frequent 1-itemsets → about  $10^7$  candidate 2-itemsets.
- **Requires repeated database scans** — one per level, so  $k + 1$  scans for a  $k$ -itemset. Very costly on disk-resident data.
- Struggles with long patterns and with low min\_sup.

The level-wise run, drawn

- H&K's 9-transaction example, min count 2:  $C_1 \rightarrow L_1$ , join to  $C_2$ , scan,  $L_2$ , join and **prune** to  $C_3$ , scan,  $L_3$ .
- Only 2 candidates survive into  $C_3$ : every other 3-itemset has an infrequent 2-subset, so it is dropped **without a scan**. That is the Apriori property at work.
- This is the shape every numerical in the companion writes out. H&K Fig 6.2.



### 4.3 FP-Growth and the FP-Tree

What is FP-Growth? Explain with example. How does it overcome Apriori's problems? 2+5 8 10 2+6

**TOP 17/23** 82 Bh, 81 Bh, 81 Ba, 80 Bh, 79 Bh, 79 Ba, 78 Bh, 75 Ch, 75 Ash, 74 Ch, 74 Ash, 73 Ch, 73 Shr, 72 Ka, 71 Shr, 70 Asa, 69 Ch **The single most-asked topic in this chapter.**

#### Key idea

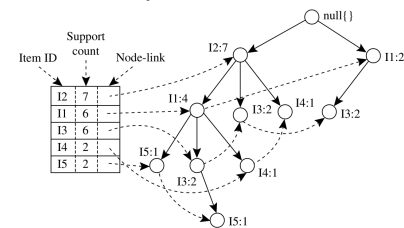
- **Generate NO candidates at all.**
- Compress  $D$  into a **frequent-pattern tree (FP-tree)**, then mine that tree by **divide and conquer**.
- Only **2 database scans** in total, against Apriori's  $k + 1$ .

#### Step 1: build the FP-tree

- Scan 1:** count every item. Discard the infrequent ones.
  - Sort the frequent items by **descending support** → the **F-list**. **Break ties consistently** (alphabetically) and say so.
  - Scan 2:** for each transaction, keep only frequent items, **reorder them into F-list order**, and insert that path into the tree.
    - Share a prefix with an existing path → just increment those counts.
    - Otherwise create new nodes with count 1.
  - Maintain a **header table**: one entry per frequent item, with a **node-link** chain threading every occurrence of that item in the tree.
- **Why descending order:** the most common items sit near the root, so paths share prefixes and the tree stays small.

#### Step 2: mine the tree (FP-Growth)

- Work **bottom-up** through the header table, ie **least** frequent item first.
- For item  $\alpha$ , follow its node-links to collect its **conditional pattern base** = every prefix path leading to  $\alpha$ , each carrying  $\alpha$ 's count at that node.
- Build the **conditional FP-tree** from that base, keeping only items that are still frequent *within* it.
- Recurse** on the conditional tree, concatenating  $\alpha$  to each pattern found.
- If a conditional tree is a **single path**, just enumerate all subsets of that path.



A finished FP-tree: **null** root, **item:count** nodes, and the header table (item, support) whose dashed **node-links** thread every occurrence of an item. Step 2 walks those links. H&K Fig 6.7.

#### Apriori vs FP-Growth (asked as a direct comparison 4 times)

|            | Apriori                   | FP-Growth                     |
|------------|---------------------------|-------------------------------|
| Candidates | Generates them            | None                          |
| DB scans   | $k + 1$                   | 2                             |
| Strategy   | Breadth-first, level-wise | Depth-first, divide & conquer |
| Structure  | Hash tree                 | <b>FP-tree</b> (prefix tree)  |
| Memory     | Low                       | Tree must fit in memory       |
| Speed      | Slower                    | Roughly an order faster       |

- **FP-Growth's weakness:** if the tree does not fit in memory, or the data is very sparse so paths barely share prefixes, the advantage disappears.

## 4.4 Handling Categorical Attributes

### How do you handle categorical attributes in data mining? Issues and fixes. 3+3 2+3+3 4

PIN 3/23 79 Bh, 73 Shr, 72 Ch

#### The problem

- Association analysis assumes **asymmetric binary** items (present / absent).
- Real tables hold categorical (Gender, Region) and continuous (Age, Income) attributes.

#### Handling: binarisation

- Convert each (attribute, value) pair into one binary item.
- Eg Gender → items Gender=Male, Gender=Female.
- Continuous attributes must be **discretised** first → Ch 2, §2.2, then binarised.

#### Issues that arise (this is what 73 Shr wants)

- **Too many items:** an attribute with many values explodes the item count. **Fix:** group rare values into an “Other” category.
- **Many items are infrequent:** rare values never reach min\_sup, so patterns involving them are lost. **Fix:** lower min\_sup, or use a per-item threshold.
- **Some items are far too frequent:** eg Gender=Male at ≈50% produces masses of trivial redundant rules. **Fix:** set a maximum support threshold too.
- **Redundancy:** values of one attribute are **mutually exclusive**, so an itemset containing Gender=Male and Gender=Female has support 0 by construction. **Fix:** never generate candidates that pair 2 values of the same attribute.
- **Loss of ordering** for ordinal attributes after binarisation.

## 4.5 Sequential, Subgraph and Infrequent Patterns

### Sequential patterns, subsequences, subgraph patterns 3+3 2 3

HOT 5/23 79 Bh, 76 Ch, 76 Ash, 73 Ch, 72 Ka Subsequence and subgraph enumeration are numericals → companion.

#### Sequential patterns

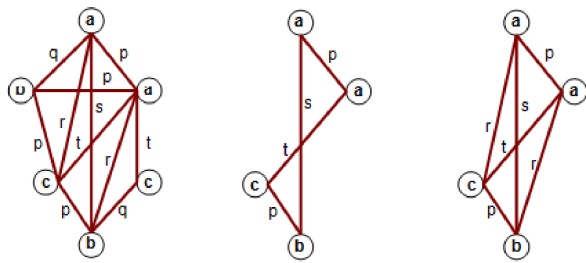
- A **sequence** is an *ordered* list of **elements**; each element is an unordered *set* of items.  $s = \langle e_1 e_2 \dots e_n \rangle$ .
- **Length** = number of *items*; a *k*-sequence has *k* items. **Not the number of elements** — this is the usual mistake.
- **Subsequence:**  $\langle a_1 \dots a_m \rangle$  is a subsequence of  $\langle b_1 \dots b_n \rangle$  if there exist  $i_1 < i_2 < \dots < i_m$  with  $a_1 \subseteq b_{i_1}, \dots, a_m \subseteq b_{i_m}$ .
  - **Order across elements must be preserved; within an element it need not be.**
- **Support** of a sequence = fraction of data sequences containing it.
- **Algorithms:** AprioriAll 76 Ash short note, GSP, SPADE, PrefixSpan. AprioriAll applies the same anti-monotone pruning to sequences.
- **Uses:** web click-streams, purchase sequences over time, DNA, telephone call patterns.

#### Subgraph patterns 79 Bh 2

- Data = a collection of **graphs** (molecules, web fragments, social networks).
- A **frequent subgraph** is one appearing in at least min\_sup of them.
- **Candidate generation by growing**, 2 schemes:
  - **Vertex growing:** join 2 graphs sharing a common subgraph, add 1 vertex.
  - **Edge growing:** join 2 graphs sharing a core, add 1 edge. **Preferred** — it generates fewer candidates.
- **The hard part is graph isomorphism:** deciding whether 2 drawings are the same graph. Handled with a canonical labelling.
- Algorithms: AGM, FSG, gSpan.

#### Infrequent / negative patterns

- A **negative pattern** is a set of items that appear together *far less* often than independence would predict.
- Eg {Coke, Pepsi} — customers buy one or the other.
- Useful for detecting substitute products and for fraud (a combination that should never occur).
- **Hard because** support is anti-monotone, so “infrequent” cannot be pruned the same way.



(a) Labeled Graph      (b) Subgraph      (c) Induced Subgraph

### Subgraph terms, drawn

(a) a labelled graph; (b) a **subgraph**: some of its vertices and edges; (c) an **induced subgraph**: some vertices and every edge between them. Candidate generation grows these one vertex or one edge at a time. Mishra's Ch 4 notes, p9.

## 4.6 PYQ Mapping

**N** = solved in the Ch 4 Numerical companion.

### 4.1 Basics

1. **TOP 7/23** Association analysis + importance in market-basket analysis. **2+5** 72 Ka **10** 72 Ch **2+6** 80 Ba
2. **PIN 2/23** Why is association analysis required? + Apriori principle w/ example. **2+6** 75 Ash, 74 Ch
3. **PIN 1/23** Importance of SUPPORT and CONFIDENCE + FP-Growth w/ example. **10** 72 Ch
4. **N PIN 1/23** Why is pattern evaluation important? + statistical interestingness measures. **8** 70 Ch
5. **PIN 1/23** Why 100% confidence is uninteresting but 99% is. **4** 80 Ba
6. **PIN 1/23** Short note: Market Basket Analysis. **4** 75 Ch

### 4.2 Apriori

7. **N** Apriori, min support count 4, confidence 80%, 7 transactions. **2+7** 80 Bh
8. **N** Candidate + large itemsets, min support 2. **4+5** 80 Ba
9. **N** Derive association rules, min sup 50%, min conf 80%. **8** 79 Bh
10. **N** Fast-food orders, min sup 2/9, min conf 7/9; frequent  $k$ -itemsets + strong rules. **5+3** 76 Ch
11. **N** L-itemsets at  $L = 3$  w/ Apriori + F-list w/ FP-Growth. **12** 76 Ash
12. **N** Candidate, frequent itemsets and rules, min sup 20%, min conf 80%. **8** 74 Ash
13. **N** Apriori w/ support 50%. **10** 70 Ch
14. **PIN 1/23** How does Apriori optimise the brute-force approach? **10** 71 Ch
15. **PIN 1/23** What are association rules? How does Apriori generate them? **10** 71 Shr

### 4.3 FP-Growth

16. **TOP 7/23** What is FP-Growth? Explain w/ example. **2+5** 80 Bh **8** 74 Ash **10** 71 Shr **2+6** 75 Ash **1+8** 72 Ka **2+5** 81 Ba
17. **N** Draw FP-tree for the given 5-transaction table. **4+4** 75 Ch
18. **N** Limitation of Apriori vs FP-Growth +  $\{M, O, N, K, E, Y\}$  database, min sup 60%, min conf 80%. **2+7** 81 Ba
19. **N** Build FP-tree transaction by transaction + mine it, threshold 33.34% / 60%. **4+4** 78 Bh
20. **N** How does FP-Growth overcome Apriori? + generate the FP-tree. **2+8** 74 Ch
21. **N** FP-tree + association rules, support 30%, confidence 75%. **10** 73 Shr
22. **PIN 1/23** Short note: FP-Tree. **5** 79 Bh

### 4.4 / 4.5 Categorical, Sequential, Subgraph

23. **N** Handle categorical attributes + generate 4 subsequences from  $\{\{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\}\}$ . **3+3** 79 Bh
24. **PIN 1/23** Categorical data, issues, how to handle them. **2+3+3** 73 Shr
25. **N** List all 4-subsequences of  $\{\{1, 3\}\{2\}\{2, 3\}\{4\}\}$  + draw candidate subgraphs by edge growing. **3+3** 76 Ch
26. **PIN 1/23** What are subgraph patterns? **2** 79 Bh
27. **PIN 3/23** Short notes: AprioriAll 76 Ash · Sequential pattern 72 Ka · Categorical data 72 Ch. **3** **4**

### Papers added by the 2026-09 merge

The six papers of DM4.1BCT.pdf (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. **verified/added**

- **N** Support and confidence + derive rules, min sup 60%, min conf 80%, 5 transactions. **2+8** 82 Bh
- **N** Apriori with **confidence 65%** and min support 50%, 4 transactions. **2+6** 81 Bh



- **N** Apriori on a  $7 \times 9$  **binary** table (min\_sup 20%) + **can any rule exist at any confidence threshold?** 5+3 79 Ba  
verified/added
- **N** Apriori in market-basket analysis, both thresholds 50%. 3+7 73 Ch
- Advantages of FP-Growth over Apriori + FP-Growth w/ example. 2+6 81 Bh
- How does FP-Growth generate frequent itemsets **without candidate generation?** 6 79 Ba
- Use of FP-Growth in market-basket analysis 73 Ch · FP-Growth in detail 70 Asa. 10 12
- What is frequent itemset mining? How do Apriori and FP-Growth optimise the brute-force approach? 15 69 Ch
- How can Apriori be used to find rules out of a frequent itemset? 7 69 Ch
- Association rules + their importance w/ example 70 Asa · short note **FP Growth** 82 Bh. 6 5
- Sequential pattern and sub-graph pattern w/ suitable example. 4+4 73 Ch

### Coverage check

- All Ch 4 PYQs covered. **14 are numerical** → companion.
- **Gaps filled by research** verified/added :
  - The **null-invariant measures** (all\_conf, max\_conf, Kulc, cosine) and the game/video counter-example for 70 Ch: your notes define only support and confidence.
  - **Issues with categorical attributes** 73 Shr: the mutual-exclusion and too-frequent-item problems are in no source note.
  - **Edge vs vertex growing** for subgraphs.

### Sources used for this chapter

- **Han & Kamber 3rd ed** ch 6 (frequent patterns, Apriori, FP-Growth, pattern evaluation) and ch 7 (advanced pattern mining).  
**Primary authority.**
- **DBK Lectures 12–14**, data mining\_/Chapter-4 Association Analysis.pdf, Associations\_Rule\_mining.ppt, FP-Growth algorithm.ppt, FP Tree.pdf, Episode\_Mining.PDF.

### Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                            | Short   | Means                         |
|----------|----------------------------------|---------|-------------------------------|
| DM       | data mining                      | DW      | data warehouse                |
| DB       | database                         | DBMS    | database management system    |
| KDD      | knowledge discovery in databases | DSS     | decision support system       |
| OLTP     | on-line transaction processing   | OLAP    | on-line analytical processing |
| ROLAP    | relational OLAP                  | MOLAP   | multidimensional OLAP         |
| HOLAP    | hybrid OLAP                      | ETL     | extract, transform, load      |
| BI       | business intelligence            | ER      | entity-relationship           |
| attr(s)  | attribute(s)                     | dim(s)  | dimension(s)                  |
| multidim | multidimensional                 | txn(s)  | transaction(s)                |
| info     | information                      | ops     | operations                    |
| repo     | repository                       | freq    | frequent / frequency          |
| config   | configuration                    | no.     | number                        |
| PCA      | principal component analysis     | SVD     | singular value decomposition  |
| SMC      | simple matching coefficient      | KNN     | $k$ -nearest neighbour        |
| ANN      | artificial neural network        | POS     | point of sale                 |
| FK       | foreign key                      | UI      | user interface                |
| GIGO     | garbage in, garbage out          | w/, w/o | with, without                 |
| rm       | remove                           | vs      | versus                        |
| Eg       | example / for example            | +Fig    | draw a figure                 |
| →        | leads to / therefore             | +       | and                           |

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth  $\geq 8$  marks. **Mark chips** such as **2+5** are the marks that question actually carried. verified/added marks anything researched and added that was not in the source notes.